

Study Guide, Exercises, and Sample Questions

CSE 4415: Wireless Networks

Islamic University of Technology (IUT)

Lecture 8.1: Understanding Vehicular Communication

1. Topics Covered

- Introduction to Vehicular Ad-Hoc Networks (VANET): definition as an application of MANET using moving vehicles as nodes; place of VANET in the Wireless Ad-hoc Network (WANET) taxonomy alongside Wireless Mesh Networks and Wireless Sensor Networks.
- V2V connectivity (each vehicle as a wireless node/router, $\approx 300\text{ m}$ – 1 km range) versus V2I connectivity (roadside infrastructure); the primary goal of road safety, plus value-added infotainment services.
- Components of VANET: On-Board Unit (OBU), Road Side Unit (RSU), and Trusted Authority (TA); the Smart Vehicle's sensing/computing elements (forward/rear radar, GPS/positioning, event data recorder, computing platform, display, communication facility).
- Layered view of a vehicle's networked ICT: local control $\rightarrow$ in-car network $\rightarrow$ driver control, extended outward via a V2V network into applications such as accident prevention, and further into congestion control, road maintenance, and end-user applications (car sharing, event management).
- Physical organization of V2V: ad-hoc links between vehicles and to RSUs over IEEE 802.11p; RSUs backhauled to the Internet over wired/wireless wide-area technology.
- Visionary and classified applications: safety applications (EEBL, PCN, RFN, LCA), convenience applications (congestion notification, route planning, parking), and commercial/infotainment applications (free-flow tolling, social networking, media exchange).
- Detailed safety mechanisms: Emergency Electronic Brake Light (EEBL), Post-Collision Notification (PCN), Lane Change Assist (LCA using radar), and Cooperative Collision Warning (CCW) built on wireless exchange rather than (or in addition to) onboard ranging sensors.
- VANET characteristics: high mobility, rapidly changing topology, predictable mobility, no power constraint (vehicle battery as an "infinite" supply), and strict time management for safety messages.
- Communication challenges: timeliness and throughput (in-vehicle), robustness and cost (across vehicles), interoperability, reachability, security, and privacy; network management issues from unbounded network size and rapidly varying signal power.
- Historical motivation: 1970s–1990s infrastructure-less prototypes (Japan CACS, Europe Prometheus, U.S. PATH) with no commercial success; the 1980s paradigm shift

to driver-advisory, infrastructure-assisted systems; the modern re-ignition via cellular tech and increased computing power.

- VANET vs. MANET vs. traditional wired networks: connection type, node mobility, and configuration; why VANET is explicitly *not* “MANET on wheels” due to different topology dynamics and communication patterns.
- Freeway vs. urban mobility models: 1D mobility with bimodal (stable/unstable) connectivity on freeways versus 2D mobility with bipolar (many/few neighbor) connectivity and obstacles in urban settings.
- Levels of infrastructure support: pure ad hoc, Stationary Support Units (SSU), Roadside Units (RSU), and Traffic Information Centers (TIC); trade-offs of infrastructure (central coordination, security, core-network load) versus no infrastructure (self-organization, low data rate).
- Convergence toward heterogeneous deployments: a single vehicle transitioning between infrastructure-supported and infrastructure-less regions during one trip.
- IEEE 802.11p fundamentals: operation without a Basic Service Set (no association/authentication), EDCA-based prioritization (from 802.11e), tolerance for high relative speeds, UTC-based timing synchronization, and 5.9 GHz channel allocation.
- Basic Safety Messages (BSM, US) and Cooperative Awareness Messages (CAM, EU) as the periodic/event-driven status broadcasts underlying V2V/V2I communication.
- WAVE/DSRC protocol family: IEEE 1609.1 (resource management), 1609.2 (security services), 1609.3 (networking services incl. WSMP), 1609.4 (multi-channel operation), and SAE J2735 (message content).
- Full WAVE/DSRC protocol stack layering against IEEE 802.11 and comparison with a conventional Internet stack (TCP/UDP/IPv6 vs. WSMP over the same MAC/PHY).
- Why plain IEEE 802.11a/b/g/n is unsuited for Car-to-X: cannot be simultaneously infrastructure and ad hoc, slow association/switching, no integral security or QoS, shared ISM spectrum, and multipath effects reducing range/speed.
- IEEE 802.11p PHY/MAC differences from 802.11a: identical 52-subcarrier OFDM structure but halved bandwidth (10 MHz vs. 20 MHz), halved subcarrier spacing (156.25 kHz vs. 312.5 kHz), doubled symbol/guard/FFT/preamble durations, halved bit rates (3–27 Mbit/s vs. 6–54 Mbit/s), and MAC extensions (random MAC address, EDCA-based QoS, multi-frequency/multi-radio capability, new ad hoc mode).
- Classic IEEE 802.11 BSS model (SSID-based logical partitioning, beacon/association/authentication) versus the new WAVE mode (wildcard BSSID, provider/user roles instead of AP/host, implied BSS membership via WAVE Service Advertisement).
- IEEE 802.11 DCF fundamentals (SIFS/PIFS/DIFS priority access, random backoff from $[0, CW]$, backoff-doubling rules) as the baseline that 802.11p’s EDCA/HCF extends.

- EDCA/HCF QoS in 802.11p: mapping 8 user priorities to 4 Access Categories (AC0–AC3), per-AC $CW_{\min}$, $CW_{\max}$, AIFSN, and TXOP limit; virtual collision handling between queues within one station; measured mean channel-access waiting time per AC and how end-to-end delay scales with node density.
- Cellular (UMTS/LTE) vs. IEEE 802.11p trade-offs, and the case for hybrid deployments combining both.
- Higher-layer channel management: dedicated 5.9 GHz DSRC band (7×10 MHz channels in the US; 5 channels in Europe), Control Channel (CCH) vs. Service Channels (SCH), and the fixed 100 ms synchronization interval with alternating 50 ms CCH/SCH slots plus guard intervals.
- WAVE Service Advertisement (WSA): broadcast on CCH to announce and describe services/data available on SCHs, enabling receivers to tune to and decode SCH traffic.
- Radio propagation for VANET links: Line-of-Sight (LOS) modeled via the Ricean distribution for small-scale fading, versus Non-Line-of-Sight (N-LOS) modeled via the Rayleigh fading distribution, with substantially larger delay spread than LOS.

2. Focus Areas

- Being able to place VANET correctly within the WANET/MANET taxonomy and explain, in your own words, why VANET is not simply “MANET on wheels.”
- Reproducing the VANET architecture diagram (OBU, RSU, TA) and correctly labeling which entity performs authentication/security, which performs local wireless relay, and which provides roadside connectivity.
- Distinguishing safety applications (few messages, small packets, high latency/reliability demands) from non-safety applications (many messages, high data rate, low latency/reliability demands) using the two-column taxonomy from the slides.
- Reading and reproducing the NHTSA/VSCC eight-application table (Lane Change Warning, Collision Warning, Emergency Brake Lights, Pre-Crash Sensing, Stop Sign Assists, Left Turn Assistance, Traffic Signal Violation, Curve Speed Warning): communication type (V2V/I2V/V2I, periodic/event-driven, P2M/P2P), frequency, latency, and range for each.
- Explaining EEBL, PCN, LCA, and CCW in terms of what problem each solves, what sensor or wireless mechanism it relies on, and how it improves on conventional (line-of-sight-only) driver awareness.
- The mapping from IEEE 802.11p PHY parameters to those of 802.11a: being able to derive doubled symbol time from halved subcarrier spacing, and explaining *why* this improves multipath/Doppler tolerance (the “highway lanes” analogy).
- Understanding the WAVE mode changes to the classic BSS model: no formal join process, wildcard BSSID usage, and the provider/user role split versus AP/host.

- The full 1609.x stack: matching each sub-standard (1609.1/.2/.3/.4/.11) to the OSI-like layer it occupies and the function it provides, and being able to redraw the protocol stack diagram (Security, Management, WSMP/IPv6/TCP/UDP, LLC, WAVE MAC, WAVE PHY).
- DCF vs. EDCA/HCF: understanding how 4 Access Categories with distinct $CW_{\min}$, $CW_{\max}$, and AIFSN produce differentiated (probabilistic) channel-access priority, and why AC_VO (safety-critical) gets the shortest AIFS and smallest contention window.
- Computing mean channel-access waiting time t_w from $AIFS[AC] = SIFS + AIFSN[AC] \times aSlotTime$ and the corresponding CW values, and interpreting the measured End-to-End Delay vs. Number-of-Nodes plot (why AC0 degrades fastest as node density grows).
- DSRC channel plan and synchronization: locating the Control Channel (Ch. 178) and Service Channels within the 5.850–5.925 GHz U.S. allocation, and computing how synchronization-interval time is split between CCH and SCH.
- The role of the WAVE Service Advertisement (WSA) as the *only* mechanism for a node to learn what is being transmitted on an SCH and how to decode it.
- Why LOS paths are modeled with a Ricean distribution while N-LOS paths are modeled with a Rayleigh distribution, and what this implies about received-signal variability and delay spread in each case.
- The cellular vs. 802.11p/ad hoc trade-off table (latency, infrastructure dependence, network load, privacy) and why a hybrid deployment is the practical answer rather than picking one exclusively.

3. How to Study These Topics

1. Redraw the VANET architecture diagram from memory (OBU-equipped vehicles, RSU, Trusted Authority) and annotate each link as V2V or V2I.
2. Build a two-column table from memory: Safety Applications vs. Non-Safety Applications, filling in message rate, packet size, latency demand, and reliability demand for each column.
3. Write one paragraph each, in your own words, explaining EEBL, PCN, LCA, and CCW — what triggers the message, who sends it, who receives it, and what action the receiver takes.
4. Derive the IEEE 802.11p PHY parameter table (bit rate, subcarrier spacing, symbol duration, guard time, FFT period, preamble duration) from the single fact “bandwidth is halved, subcarrier count stays at 52,” and check every value against the slide table.
5. Redraw the WAVE/1609.x protocol stack (Security & Management columns; WSMP/IPv6/TCP-UDP; LLC; WAVE MAC with 1609.4 multi-channel coordination; WAVE PHY) and label each 1609.x sub-standard next to the layer it governs.

6. Prepare a one-page handwritten comparison: IEEE 802.11 DCF (single queue, DIFS, binary exponential backoff) vs. IEEE 802.11p EDCA/HCF (4 ACs, AIFS, per-AC CW, TXOP, internal virtual collision).
7. Practice computing $AIFS[AC]$ and approximate t_w for each Access Category using the sample parameter table (SlotTime = $13\mu s$, SIFS = $32\mu s$, $CW_{\min} = 15$, $CW_{\max} = 1023$), and compare your numbers to the slide's t_w column (264, 152, 72, $56\mu s$).
8. Draw the DSRC channel-switching timeline (100 ms sync interval, alternating 50 ms CCH/50 ms SCH, small guard intervals) and mark where BSM broadcasts happen versus where WSA/WSM/IP data happen.
9. Summarize, in one sentence each, the five levels of infrastructure support (pure ad hoc, SSU, RSU, TIC) and the trade-off diagram (Infrastructure $\Leftrightarrow$ No Infrastructure).
10. Write a short paragraph contrasting Ricean (LOS) and Rayleigh (N-LOS) fading in your own words, including which one has larger delay spread and why that matters for VANET safety-message reliability.

4. Exercises

4.1. VANET Fundamentals and Architecture

1. Define VANET in one sentence and place it correctly in the WANET taxonomy (relative to Wireless Mesh Networks, MANET, and Wireless Sensor Networks).
2. List the three major components of a VANET and, for each, give its primary responsibility (communication relay, roadside connectivity, or security/authentication).
3. A vehicle's OBU wants to (a) exchange status with a car 500 m ahead in the same lane, and (b) obtain Internet access near an intersection. Which VANET component handles case (a) and which handles case (b)? Explain briefly.
4. Describe, in order, the flow of a message from a vehicle's local control systems (steering, brakes, throttle) up to a V2V-based accident-prevention application. Name each layer in the flow.
5. The lecture states typical V2V range is "approximately 300 m–1 km." If a vehicle is traveling at 100 km/h and another vehicle is 800 m ahead traveling in the same direction at 100 km/h, how many seconds would it take for the gap to close to 300 m if the following vehicle were somehow approaching at a *relative* speed of 20 km/h? (Convert units carefully.)
6. List the five components of a "Smart Vehicle" shown in the lecture (radar, positioning, computing, display, communication, event recorder) and match each to the safety or convenience function it primarily supports.

4.2. V2X Communication Patterns and Taxonomy of Use Cases (with Numericals)

7. Name the three communication patterns illustrated under Vehicle-to-X (V2X): "...to vehicle," "...to infrastructure," and "...to home." For each, give one concrete VANET example from the lecture.

8. Using the Taxonomy-of-Use-Cases diagram, classify each of the following as Non-Safety/Comfort, Non-Safety/Traffic-Information, Safety/Situation-Awareness, or Safety/Warning-Message: (a) Adaptive Cruise Control, (b) Congestion Information, (c) Traffic Light Violation warning, (d) In-vehicle entertainment.
9. The lecture states that Non-Safety applications need *many messages, high data rate, low latency demands, low reliability demands*, while Safety applications need the opposite profile. Explain in one paragraph why a safety message can tolerate lower *reliability per single transmission* yet the overall safety function still ends up highly reliable. (Hint: think about message repetition rate.)
10. Using the Diversity-of-Use-Cases table (Hazard warning 250 m/10 s/All; City-wide alarm 20 km/hours/All; Travel-time information 5 km/minutes/All), compute how many hazard-warning messages a single vehicle would need to (re-)broadcast in a 2-minute window if the message is re-sent once every 10 s while the hazard remains relevant.

4.3. Safety Applications: EEBL, PCN, LCA, CCW (with Numericals)

11. Explain how EEBL overcomes the visibility limitations of conventional brake lights. Name two additional pieces of information (beyond “braking is happening”) that an EEBL Basic Safety Message can carry.
12. A truck brakes hard on a foggy road. Car A (following the truck directly) and Car B (two vehicles behind, with no direct line of sight to the truck) both use EEBL. Explain step by step how Car B still receives a timely warning.
13. Describe what triggers a Post-Collision Notification (PCN) and what two vehicle-state variables are used to detect the triggering event.
14. Lane Change Assist (LCA) uses radar to monitor blind spots and traffic behind the vehicle, and measures “closing speed.” If a vehicle in the adjacent lane is 40 m behind and closing at a relative speed of 20 km/h, roughly how many seconds does the LCA system have before that vehicle reaches the driver’s position? (Convert 20 km/h to m/s first, then divide distance by relative speed.)
15. Explain the key difference between a traditional (sensor-only: radar/lidar/camera) Cooperative Collision Warning (CCW) system and a wireless-based CCW system. Why can wireless-based CCW build awareness of vehicles that traditional sensors cannot directly detect (e.g., vehicles two cars ahead)?
16. The lecture states CCW “calculates the relative safety distance between vehicles” and stores data in a Motor Vehicle Event Data Recorder (MVEDR). Why is this storage useful *after* an accident has already happened, even though CCW’s main purpose is prevention?

4.4. IEEE 802.11p PHY/MAC and Differences from 802.11a (with Numericals)

17. Starting from IEEE 802.11a’s 20 MHz channel bandwidth and 52 subcarriers, compute the subcarrier spacing. Then compute the 802.11p subcarrier spacing for a 10 MHz channel with the same 52 subcarriers. Confirm your answers against the values 312.5 kHz and 156.25 kHz.

18. Symbol duration is the reciprocal of subcarrier spacing. Using your answer to the previous question, compute the OFDM symbol duration for 802.11a and for 802.11p, and confirm they match $4\mu\text{s}$ and $8\mu\text{s}$ respectively.
19. Given that guard time, FFT period, and preamble duration all double from 802.11a to 802.11p ($0.8\mu\text{s} \rightarrow 1.6\mu\text{s}$, $3.2\mu\text{s} \rightarrow 6.4\mu\text{s}$, $16\mu\text{s} \rightarrow 32\mu\text{s}$), explain in one sentence *why* doubling these quantities improves robustness to multipath delay spread in a vehicular environment.
20. IEEE 802.11a supports bit rates $\{6, 9, 12, 18, 24, 36, 48, 54\}$ Mbit/s. IEEE 802.11p (half-clocked mode) supports exactly half of each of these values. List the resulting 802.11p bit-rate set and confirm it matches $\{3, 4.5, 6, 9, 12, 18, 24, 27\}$ Mbit/s.
21. A 300-byte Basic Safety Message is sent at the lowest 802.11p data rate (3 Mbit/s). Compute the raw transmission time using $T = \frac{L \times 8}{R}$, and compare it against the same message sent at the highest rate (27 Mbit/s).
22. List at least four reasons, from the lecture, why plain IEEE 802.11{a,b,g,n} (as opposed to 802.11p) is unsuitable for Car-to-X communication.
23. Explain the difference between the classic IEEE 802.11 BSS model (SSID-based, explicit join via beacon/authentication/association) and the new WAVE mode (wildcard BSSID, implied membership via WAVE Service Advertisement). Why does WAVE mode avoid the overhead of formal joining?

4.5. EDCA/HCF QoS in IEEE 802.11p (with Numericals)

24. List the four Access Categories (AC0–AC3, or equivalently AC_BK, AC_BE, AC_VI, AC_VO) in order of increasing priority, and state which one is used for safety-critical messages such as BSMs.
25. Using the sample parameter table ($\text{SlotTime} = 13\mu\text{s}$, $\text{SIFS} = 32\mu\text{s}$, $CW_{\min} = 15$, $CW_{\max} = 1023$) and the AIFSN values $\{9, 6, 3, 2\}$ for AC0–AC3, compute $\text{AIFS}[AC] = \text{SIFS} + \text{AIFSN}[AC] \times \text{SlotTime}$ for each Access Category.
26. Using the CW formulas from the lecture table ($CW_{\min}^{AC-BK} = CW_{\min}$, $CW_{\min}^{AC-VI} = \lceil (CW_{\min} + 1)/2 \rceil - 1$, $CW_{\min}^{AC-VO} = \lceil (CW_{\min} + 1)/4 \rceil - 1$), compute $CW_{\min}$ for AC-VI and AC-VO given $CW_{\min} = 15$, and verify they equal 7 and 3 respectively.
27. Explain what a “virtual collision” is in EDCA/HCF and why it can occur even though only one radio is transmitting per station.
28. Given the measured t_w values (264, 152, 72, $56\mu\text{s}$ for AC0–AC3), explain why AC0 (background/lowest priority) has by far the largest mean channel-access waiting time, and connect this to the End-to-End-Delay-vs-Number-of-Nodes plot showing AC0 degrading fastest as node count grows from 50 to 300.
29. A station has AC_VO backoff counter at 0 when a higher-priority external event (channel becomes busy) occurs. Walk through, step by step (as in the lecture’s channel-access state diagrams), what happens to the AC_VO queue’s backoff counter when the channel later becomes idle again for AIFS.

4.6. WAVE/DSRC Channel Management and Protocol Stack (with Numericals)

30. Name the five components of the IEEE 1609.x family covered in the lecture (1609.1, 1609.2, 1609.3, 1609.4, 1609.11) and state, in a few words, what each one governs.
31. The U.S. DSRC allocation reserves 7 channels of 10 MHz each in the 5.850–5.925 GHz band. Compute the total spectrum (in MHz) this represents, and compare it to the European ETSI allocation of 5 channels of 10 MHz each.
32. Using the DSRC channel-switching timing diagram (100 ms synchronization interval, alternating 50 ms on CCH and 50 ms on SCH), compute how many full synchronization intervals occur in 1 second, and how many milliseconds per second a single-radio device spends listening on the Control Channel.
33. A vehicle wants to receive a Traffic Information Service being broadcast on a Service Channel (SCH). Explain, step by step, how the WAVE Service Advertisement (WSA) mechanism allows the vehicle to discover and correctly decode this SCH traffic, referencing the Node-A/Node-B sequence diagram from the lecture.
34. Basic Safety Messages (BSM) are sent on the Control Channel (CCH) while WAVE Short Messages (WSM) and IP data are sent on Service Channels (SCH). Explain why safety-critical messages are kept on a single, dedicated, always-monitored channel rather than being distributed across the SCHs along with data traffic.
35. Draw (in words) the full WAVE protocol stack from PHY up to Application, labeling which 1609.x standard governs each layer, and where WSMP sits relative to a conventional TCP/UDP/IPv6 stack.

4.7. Cellular vs. 802.11p, Challenges, and Propagation Models

36. List two advantages and two disadvantages of using UMTS/LTE (cellular) for vehicular communication, and two advantages and two disadvantages of using IEEE 802.11p/ad hoc instead.
37. Explain, with reference to the lecture's four challenge categories (Communication, Networking, Mobility, Security), why VANET security is harder than in a typical MANET (hint: no reliable uplink to a central authority, heterogeneous user base).
38. Distinguish between a Line-of-Sight (LOS) and a Non-Line-of-Sight (N-LOS) vehicular radio link. Which fading distribution (Ricean or Rayleigh) is used to model each, and which of the two typically has the larger delay spread?
39. Two vehicles are approaching a curve: Vehicle 1 travels at 40 km/h and Vehicle 2 (around the bend, out of direct line of sight, at a truck blocking the path) travels at 0 km/h, separated by 200 m of curved road. Would you model this V2V link as LOS or N-LOS, and which fading distribution applies? Justify your answer using the scenario description from the lecture.
40. Summarize, in your own words, why the lecture concludes that a *hybrid* system (both cellular and 802.11p deployed together) is the practical solution, rather than choosing one technology exclusively.

5. Sample Questions

5.1. Short Questions

1. What does VANET stand for, and of which broader network category is it an application?
2. Name the three major components of a VANET architecture.
3. What is the typical V2V communication range stated in the lecture?
4. What is the primary goal of VANET, and what additional (secondary) services can it support?
5. What does OBU stand for, and what is its main role?
6. What does RSU stand for, and where is it typically deployed?
7. What does TA stand for, and what is its main responsibility?
8. Name the five characteristics of VANET listed in the lecture.
9. What is EEBL, and what problem does it solve that conventional brake lights cannot?
10. What is the difference between a traditional (sensor-based) and a wireless-based Co-operative Collision Warning system?
11. How many OFDM subcarriers does IEEE 802.11p use, and how does its channel bandwidth compare to IEEE 802.11a?
12. Name the four Access Categories used in IEEE 802.11p's EDCA/HCF QoS scheme, from lowest to highest priority.
13. What is a WAVE Service Advertisement (WSA), and on which channel is it broadcast?
14. How long is one DSRC synchronization interval, and how is it split between CCH and SCH?
15. Which fading distribution models LOS vehicular links, and which models N-LOS links?

5.2. Descriptive Questions

16. Describe the complete VANET architecture (OBU, RSU, TA) and explain how a message travels from one vehicle, through infrastructure, to a Trusted Authority for authentication.
17. Explain, with examples, the difference between Safety and Non-Safety use cases in the VANET taxonomy, covering message frequency, packet size, latency demands, and reliability demands for each.
18. Describe the four safety mechanisms covered in the lecture — EEBL, PCN, LCA, and CCW — explaining for each what triggers it, what information is exchanged, and what benefit it provides over conventional (non-connected) driving.

19. Explain, step by step, how IEEE 802.11p differs from IEEE 802.11a at the PHY layer (bandwidth, subcarrier spacing, symbol duration, bit rate) and at the MAC layer (random MAC address, EDCA-based QoS, ad hoc mode), and explain why each PHY change improves vehicular robustness.
20. Describe the full WAVE/DSRC protocol stack (1609.1 through 1609.4, plus 1609.11) and explain how it sits on top of IEEE 802.11p, contrasting WSMP with a conventional TCP/UDP/IPv6 path.
21. Explain the DSRC channel management scheme: the Control Channel vs. Service Channels, the 100 ms synchronization interval, guard intervals, and the role of the WAVE Service Advertisement in making SCH traffic discoverable.

5.3. Analysis and Design Questions

22. A city wants to deploy VANET-based Emergency Brake Light Warning and Curve Speed Warning applications city-wide. Using the NHTSA application table (frequency, latency, range for each), discuss what differs in the network design (message rate, coverage, whether V2V or I2V is required) between these two applications, and propose a channel/priority (Access Category) assignment for each.
23. Compare cellular (UMTS/LTE) and IEEE 802.11p deployments for a highway safety-warning system spanning a 50 km stretch of road with no existing RSU infrastructure. Discuss latency, infrastructure cost, coverage, and propose whether a hybrid approach would be preferable, and why.
24. A vehicle is transitioning from a city center (RSU-covered) onto a rural highway (no RSU coverage) and then into a tunnel (temporary complete signal loss, LOS blocked). For each of the three phases, state which V2X communication pattern(s) remain available, and what propagation model (LOS/N-LOS, Ricean/Rayleigh) best describes the vehicle's V2V links in each phase.
25. Design a channel-access priority scheme (using the 4 EDCA Access Categories) for a vehicle that simultaneously needs to send: (a) a periodic 10 Hz BSM, (b) an event-driven Emergency Brake Light message, (c) an infotainment audio stream, and (d) a background software-update download. Justify your AC assignment for each traffic type using the $CW_{\min}/CW_{\max}/AIFS_N$ trade-offs from the lecture.
26. Using the U.S. DSRC channel plan (Ch. 172 Collision Avoidance Safety, Ch. 178 Control Channel, Ch. 184 Dedicated Public Safety, and shared public-safety/private-service channels in between), propose which channel(s) a newly-deployed pedestrian-crossing warning application (safety-critical, low data rate) should use, and justify your choice against the channel plan's design intent.
27. Discuss why the lecture emphasizes that VANET is *not* simply "MANET on wheels." Identify at least three specific differences (mobility model, topology dynamics, communication patterns, or infrastructure availability) and explain the practical protocol-design consequence of each.